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Tutorial 4: Pick and Place in MuJoCo with Inverse Kinematics

Off-the-shelf Mink vs. closed-loop Damped Least Squares vs. constrained QP

Anirudha Pratihar (Roll No. 26210015)

Code: https://github.com/Anirudha-Pratihar/IK-Assignment-Robotics Mink: https://github.com/kevinzakka/mink

Abstract

We build a randomized pick-and-place pipeline for a 6-DOF HEAL arm in MuJoCo (cube → tray) and compare three differential-IK solvers on identical episode seeds: the Mink library, a closed-loop damped-least-squares (DLS) controller, and a custom closed-loop quadratic program (QP) with joint-position, velocity and collision constraints. The task and dexterous workspaces are computed by IK sampling and used to size the table. On the designed table Mink, DLS and QP reach 25/25, 25/25 and 25/25 respectively; the methods differ in cost and, under stress, in constraint handling: DLS needs 24 instead of 94 iterations per episode but violates the joint-velocity limit in 446 control ticks in a fast-motion test, while QP and Mink do not (0/0).

1 Introduction and goals

The goal is a complete pick-and-place pipeline driven by inverse kinematics (IK): analyse the workspace, design the table, randomize the cube pose (x, y, ψ), execute approach–grasp–lift–transit–place, log every success and failure with a reason, and compare IK methods under a fixed protocol. Concretely we (i) compute task and dexterous workspaces, (ii) design $L \times W \times H$ of the table from them, (iii) run N=25 randomized episodes per method with identical seeds, (iv) categorise failures and test mitigations, and (v) compare Mink, DLS and QP with joint-limit and velocity-violation counts.

2 Methods

2.1 Robot, gripper and simulation

The arm is the 6-DOF HEAL model from the lab folder (base at the origin, on the floor). The raw torque motors are replaced by joint position servos (k_p=1500, k_d=80) with feed-forward of gravity ($q_{\text{frc,bias}}$) and of the joints' passive damping (55 N m s/rad), integrated with MuJoCo's implicit-fast integrator (Δt=2 ms). A parallel-jaw gripper is attached at the flange site; the TCP is the pad centre. Grasping is physical (friction, elliptic cones): the cube (4 cm, 50 g, μ=1.2) is held by pad contact only. [TODO: state which gripper produced the final results: Robotiq 2F-85 or the built-in fallback (the reference runs here used the fallback with the same interface)]

Franka Panda (7-DOF). The same code runs on the Franka Panda with its built-in two-finger hand (kept servos; stiffer finger gain). The arm is redundant, so every solver gets a null-space posture term towards the ready pose (DLS: null-space projector; QP: posture term in the cost; Mink: low-cost `PostureTask`). The base is mounted 0.8 m above the floor, so the workspace analysis is repeated for this robot (table: H=0.75 m, $L \times W$=0.30×0.72 m).

2.2 Workspace analysis and table design

For each TCP position on a grid (5 cm) and several heights we solve IK for a *top-down* gripper pose. **Task workspace**: positions reachable for at least one yaw. **Dexterous workspace**: positions reachable for *all* sampled yaws in $[-\pi, \pi)$ (continuation over yaw, joint limits enforced, kinematics only). For each candidate height H the masks at the grasp (H+3 cm), pre-grasp and carry heights are intersected, and the largest all-feasible rectangle (histogram method) defines $L \times W$; the H with the largest area wins (Fig. **??**, Tab. **??**). The tray sits on the table; the cube spawns in the remaining area with a margin to the tray.

2.3 Episodes and pipeline

Seed s gives cube (x, y) uniform in the spawn region and $\psi \sim U[-\pi, \pi]$, *identically for every method.* The square cube allows four grasp yaws $\psi + k\pi/2$; the one closest to the current tool yaw is tried first, and the planner can fall back to the others (`grasp_yaw_retries`). Phases: *approach* (above cube) → *descend* → close → *lift* → *transit* (above tray) → *place* → open → *retreat* → verify (cube inside the tray and at rest). Each phase is a straight TCP line with minimum-jerk timing, tracked by the chosen IK method on a private copy of the model (Planner), and replayed by the servo in the physics simulation (Controller); the Env/Planner/Controller split matches the tutorial's sequence diagram. Approach and descend are dry-run first so alternative yaws can be tried.

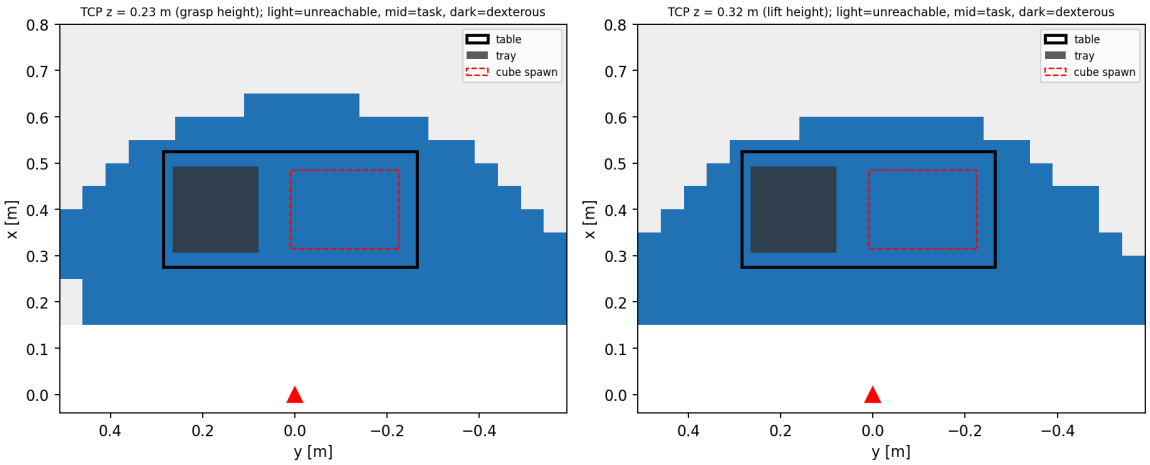


Figure 1: Task (mid) and dexterous (dark) workspace at grasp and carry height, with the designed table, tray and cube-spawn region. The workspace is a ring: the tool-down pose cannot reach close to the base at low height.

Table 1: Table design.

Parameter	Symbol	Value	Rationale
Table length	L	0.25 m	fits the dexterous x-range at grasp and lift heights
Table width	W	0.55 m	fits the dexterous y-range
Table height	H	0.20 m	maximises dexterous area; vertical approach within joint limits

2.4 IK methods

All three share the same closed-loop task law: each tick (Δt=20 ms) they close a fraction κ=0.8 of the 6-D pose error, $v^\star = \kappa\, e/\Delta t$, so differences come from the solver, not the gain.

- **DLS:** $\dot{q} = J^\top (JJ^\top + \lambda^2 I)^{-1} v^\star$, λ=0.03 with adaptive growth near $\sigma_{\min}(J) \to 0$. Limits: position clipping only; collision: potential-field repulsion along contact normals, active within a per-group band above the keep-out distance.
- **QP:** $\min_{\dot{q}} \ \frac{1}{2}\|J\dot{q} - v^\star\|^2 + \frac{\mu}{2}\|\dot{q}\|^2$ s.t. $\max(-\dot{q}_{\max}, \frac{q_{\min}-q}{\Delta t}) \le \dot{q} \le \min(\dot{q}_{\max}, \frac{q_{\max}-q}{\Delta t})$ and, for every geom pair closer than 6 cm, $d + \mathbf{n}^\top (J_2 - J_1)\dot{q}\,\Delta t \ge d_{\min}$ (signed distance from `mj_geomDistance`); solved with DAQP.
- **Mink:** `FrameTask` on the TCP with `ConfigurationLimit`, `VelocityLimit` and one `CollisionAvoidanceLimit` per geom group. Mink is the off-the-shelf baseline; it solves essentially the same QP as ours (same bounds, same distance constraints), differing in how the task and limits are assembled.

Collision handling covers arm and hand versus table and tray, pads versus cube (pre-grasp), and a cube-sized geom welded to the hand versus the tray and table while carrying. **Smart carry height:** independently of the reactive constraints, the planner lifts to at least $h_{\text{wall}} + c + h_{\text{grasp}}$ so the carried cube passes above the tray wall.

2.5 Metrics and protocol

Per episode we log: success, failure reason and phase, solver time (wall time inside the IK solvers), IK iterations (sum over waypoints of the iterations a fixed-target solve needs to converge), minimum signed clearance (arm/hand vs. table+tray, and cube vs. tray while carrying), joint-limit events (raw update left the range), velocity events ($|\dot{q}| > \dot{q}_{\max}$=2 rad/s unless stated), TCP tracking error and placement error. Success rates carry Wilson 95% intervals; since seeds are shared, methods are compared with paired tests (exact McNemar for success, Wilcoxon signed-rank for continuous metrics). For the combined Franka set each seed is run by only one method, so Fisher's exact test and Mann–Whitney U are used instead.

3 Results

Nominal. Success is 25/25, 25/25, 25/25 (Mink, DLS, QP; Tab. **??**); solve times are 154/120/162 ms (Mink/DLS/QP) per episode and IK iterations 94/24/94. DLS converges in fewer iterations because it has no velocity bound, whereas Mink and QP are limited to $\dot{q}_{\max}$; the paired tests confirm DLS differs significantly from QP in time and iterations while Mink and QP are statistically indistinguishable (same optimisation problem). Minimum arm clearance is 37 mm and cube–tray clearance 18 mm for all methods; placement error is 4.1 mm.

Table 2: Nominal experiment (designed table, 25 episodes, same seeds).

Method	Succ.	Rate [%] (95% CI)	Solve [ms]	IK it.	Min clr [mm]	Cube–tray [mm]	Vel. viol.	Limit ev.	Place err [mm]
Mink	25/25	100 (87–100)	154±19	94±12	37	18	0	0	4.1
DLS	25/25	100 (87–100)	120±17	24±1	37	18	0	0	4.1
QP	25/25	100 (87–100)	162±27	94±12	37	18	0	0	4.1

Pair	only 1st ok	only 2nd ok	p success	p solve time	p IK it.
Mink vs DLS	0	0	1.000	<0.001	<0.001
Mink vs QP	0	0	1.000	0.120	0.655
DLS vs QP	0	0	1.000	<0.001	<0.001

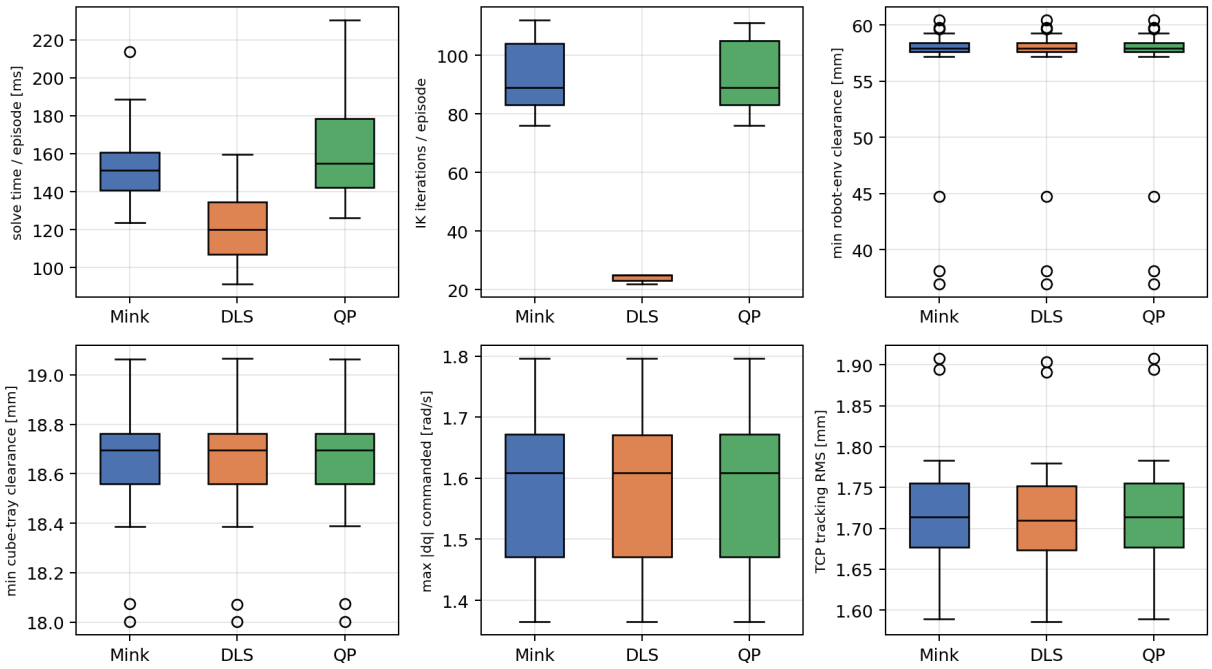


Figure 2: Nominal experiment: distribution over 25 episodes of solver time, IK iterations, clearances, peak commanded joint velocity and tracking error.

Stress and constraints. Under stress (Tab. **??**) success drops to 16/25, 15/25, 16/25 (Mink, DLS, QP). In this run DLS reports 737 joint-limit events and 726 velocity violations, while Mink and QP report none because both are enforced inside the QP. In the fast-motion test DLS commands up to 2.55 rad/s (limit 1.5) in 446 ticks and loses one cube in transit; Mink/QP stay at the limit but pay with a larger tracking error (4.3 mm vs. 3.1 mm) and placement error (11.3 mm).

Franka Panda (combined 25 episodes). Success is 9/9, 8/8, 8/8 (Mink, DLS, QP; Tab. **??**). The cost pattern matches the HEAL arm: DLS needs 21 IK iterations against 65 (Mink) and 64 (QP) because it has no velocity bound, with solve times 177/154/155 ms; clearances (37 mm arm, 20 mm cube–tray) and placement error (0.5 mm) are equal across methods. With only 8–9 episodes per method, success rates cannot be separated (Wilson 95% lower bound about 68%); the remaining 16/17/17 episodes complete 25 per method and make the comparison paired.

4 Failure analysis and mitigations

Categories. (1) *IK infeasible / limits*: the dominant failure under stress; the cube lies outside the dexterous region (Fig. **??**a,b), so no method can succeed and the failure sets of the three methods are nearly identical. (2) *Collision*: without avoidance every episode with the low carry height clips the tray wall (0/25 success); with reactive constraints only, Mink and QP avoid the wall (min. cube–tray clearance 44 mm) but stall in front of it (0/25 success, Fig. **??**c) because a local constraint cannot choose to go *over* the wall, and DLS's repulsion is too weak (3/25, collisions). (3) *Grasp slip*: caused by aggressive velocities (Fig. **??**d). (4) *Place miss / grasp miss*: none occurred in the nominal runs.

Mitigations tested. *Alternate grasp yaw* raises stress success from 16/25, 15/25, 16/25 to 17/25, 16/25, 17/25 at a higher solve time (202 → 342 ms for Mink); the gain is small because the remaining failures are all at joint limits or out of reach, which a different wrist yaw cannot always fix. *Geometry-aware carry height* combined with reactive constraints gives 25/25, 25/25, 25/25 (Mink, DLS, QP) in the same low-carry setting. Further options: better pre-grasp heights, a different approach direction, larger collision margins, a table placed inside the dexterous

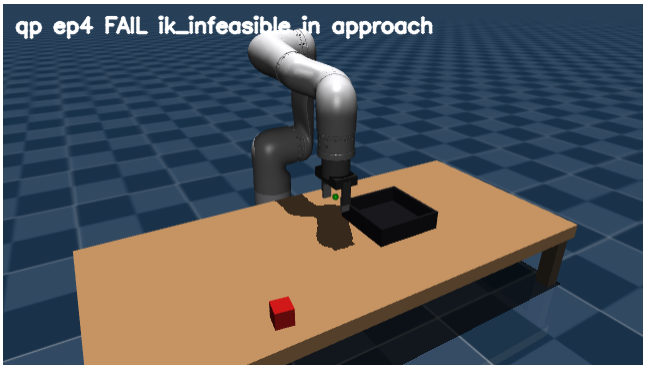
Table 3: Success (of 25) across experiments. *stress*: table 2.2× larger (partly unreachable), cubes next to the tray, joint ranges at 55%; *retry4*: stress + up to 4 grasp yaws; *lowcarry*: carry height 7.5 cm (cube would clip the tray wall) without avoidance / reactive avoidance only / reactive + smart carry height; *fast*: v_{cart}=0.9 m/s, $\dot{q}_{\max}$=1.5 rad/s.

Experiment	Mink	DLS	QP
main	25/25	25/25	25/25
stress	16/25	15/25	16/25
stress_retry4	17/25	16/25	17/25
lowcarry_noCA	0/25	0/25	0/25
lowcarry_ca	0/25	3/25	0/25
lowcarry_smart	25/25	25/25	25/25
fast	25/25	24/25	25/25

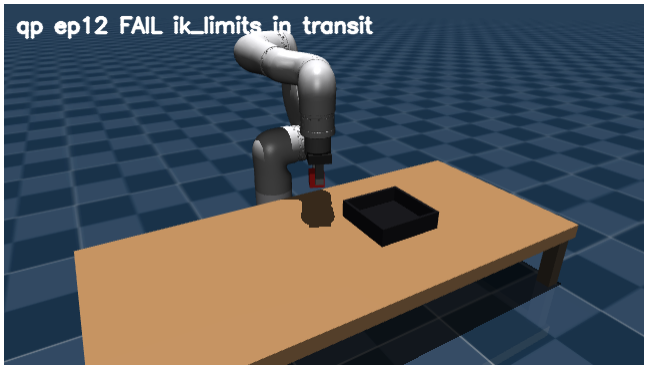
Table 4: Franka Panda 7-DOF, combined set: 25 seeds shared out round-robin (9/8/8 episodes), so the comparison is unpaired.

Method	Succ.	Rate [%] (95% CI)	Solve [ms]	IK it.	Min clr [mm]	Cube–tray [mm]	Vel. viol.	Limit ev.	Place err [mm]
Mink	9/9	100 (70–100)	177±11	65±9	37	20	0	0	0.5
DLS	8/8	100 (68–100)	154±53	21±1	37	20	0	0	0.4
QP	8/8	100 (68–100)	155±21	64±11	37	20	0	0	0.5

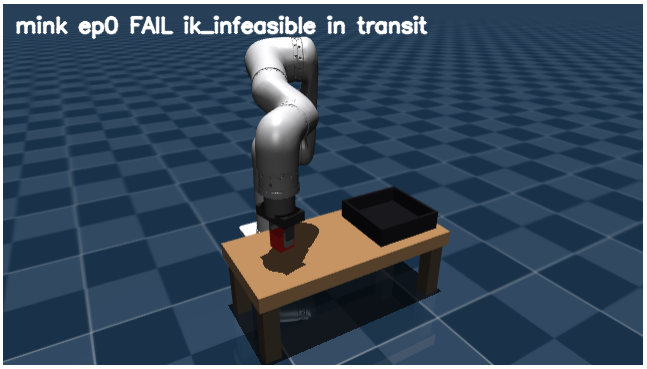
Pair	test	n (1st/2nd)	p success	p solve time	p IK it.
Mink vs DLS	unpaired	9/8	1.000	0.200	<0.001
Mink vs QP	unpaired	9/8	1.000	0.036	1.000
DLS vs QP	unpaired	8/8	1.000	0.645	<0.001



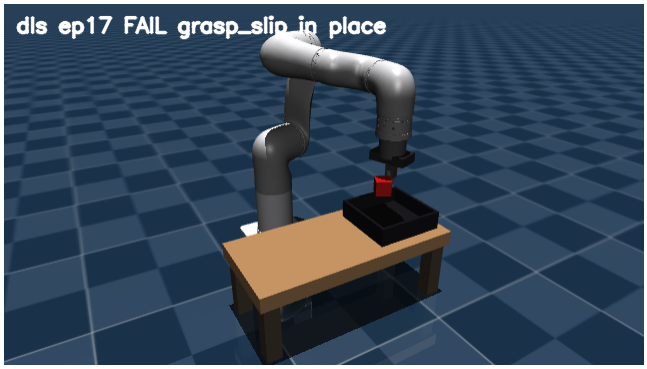
(a) Unreachable pre-grasp (cube too far).



(b) Joint limit reached during transit.



(c) Reactive avoidance only: stalls before the tray wall.



(d) DLS, fast motion: cube slips out.

Figure 3: Representative failures (screenshots taken at the moment of failure).

region (Sec. 2.2).

5 Discussion and conclusions

Constraint handling, not success rate, separates the methods on this task: on a well-designed table all three succeed, and under stress the dominant failures (unreachable cubes) are workspace-limited and shared by every solver; the workspace analysis and the table design therefore matter more than the choice of IK method. Where the methods do differ, QP/Mink guarantee limits and velocity bounds and DLS does not, at the price of more iterations and solver time. Reactive collision constraints are necessary but not sufficient; they must be combined with geometry-aware

Table 5: Failure reasons in the stress experiment (counts over 25 episodes).

Method	ik_infeasible	ik_limits
Mink	5	4
DLS	6	4
QP	5	4

Table 6: What each method handles better (supported by the experiments above).

	Strengths	Weaknesses	Evidence
Mink	fast to integrate; limits and collision built in; same behaviour as our QP	less control over constraint formulation; extra dependency; tunable gain semantics (per-step fraction)	identical outcomes to QP in every experiment
DLS	simplest; fewest iterations; robust near singularities	no velocity/limit guarantee; repulsion needs per-group tuning and was insufficient for the tray wall	446 velocity violations (fast); 3/25 (low carry, CA only)
QP	constraints enforced exactly (position, velocity, distance); graceful on infeasible targets	needs a solver; solve time higher; reactive only—cannot plan around obstacles	no velocity/limit events in our runs (0 in the fast test); stalls at the wall without smart carry height

waypoints (here: without them, 0/25 succeed at the low carry height). **Limitations:** 25 episodes per cell give wide intervals (Wilson 95% $\approx$ 87–100% at 25/25); IK planning is kinematic and replayed through a servo; collisions are checked on collision meshes only; results depend on the gripper model. [TODO: add observations from your own video trials and final gripper]

References

[1] E. Todorov, T. Erez, Y. Tassa. MuJoCo: a physics engine for model-based control. IROS 2012.

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[3] Y. Nakamura, H. Hanafusa. Inverse kinematic solutions with singularity robustness. J. Dyn. Syst. Meas. Control, 1986.

[4] D. Arnström et al. A dual active-set solver for embedded QP (DAQP). IEEE TAC, 2022.